

D850

General Biology I Lab

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A student group working in a well-equipped teaching lab with new digital balances wants to study whether light color affects the growth of radish seedlings. Their instructor, who has taught the course for years, asks them to write a hypothesis before ordering seeds. Which statement is a testable hypothesis?

- A. Blue light will always produce the tallest plants of any color.
- B. If seedlings are grown under blue light instead of red light, their mean height after 14 days will be greater.
- C. Does light color affect radish seedling growth?
- D. Radish seedlings prefer blue light.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

In a lab run in a room kept at 22 °C, a team places equal-sized pieces of potato in salt solutions of 0%, 2%, 4%, and 6%. Each piece soaks for 30 minutes. Using a scale they calibrated that morning, the team records each piece's change in mass. What is the dependent variable?

- A. The change in mass of each potato piece.
- B. The 30-minute soaking time.
- C. The room temperature of 22 °C.
- D. The salt concentration of the solution.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

A researcher measures how fast yeast produces carbon dioxide at 20, 30, and 40 °C, using the same amount of sugar and yeast in each flask. What is the independent variable?

- A. Temperature.
- B. The yeast, because it is the organism being studied.
- C. The amount of sugar in each flask.
- D. The rate of carbon dioxide production.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A lab team tests whether a new liquid fertilizer increases bean plant mass. They buy seeds from a well-known supplier and grow 20 plants in each group. The treatment group gets fertilizer dissolved in 50 mL of water every 3 days. The team's lab notebook is neat and fully dated. What should the control group receive?

- A. Seeds from a different supplier, to check seed quality.
- B. Twice the fertilizer, to see a stronger effect.
- C. 50 mL of plain water every 3 days, with all other conditions the same.
- D. Nothing at all — no water and no fertilizer.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

A student tests an antibacterial soap. She washes one set of countertop swab plates with soap and warm water and leaves a second set completely unwashed. The soap group grows fewer colonies. Why can't she conclude the soap itself reduced bacteria?

- A. Colony counts cannot measure bacteria.
- B. A negative control is never valid in microbiology.
- C. The dependent variable should have been the soap.
- D. The groups differ in both soap and water-washing, so the soap's effect cannot be separated.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A student with an interest in agriculture tests a plant hormone using one treated tomato plant and one untreated tomato plant on a sunny windowsill. After three weeks the treated plant is 12 cm taller. The student's poster uses clear, colorful graphs. What is the most serious weakness of this study?

- A. The graphs should have been in black and white.
- B. Windowsill light makes plant studies invalid in every case.
- C. A 12 cm difference is too small to matter.
- D. There is only one plant per group, so the difference could be due to chance.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A research assistant in a grant-funded lab records enzyme activity in 10 trials. Nine trials fall between 42 and 48 units; one reads 71 units. The lab director hopes to publish soon, and the deadline is Friday. No exclusion rule was written in the protocol. What should the assistant do?

- A. Report the 71 as the main finding because it is the most striking.
- B. Submit the nine values by Friday and add the 71 later if asked.
- C. Report all 10 values, flag the 71, and look for a cause (and repeat the trial) rather than deleting it.
- D. Delete the 71 since it does not fit the other nine.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A simulated lab report on catalase activity is 6 pages long and was written by a team of four. It contains this sentence: "Enzyme activity likely fell at 60 °C because the high heat changed the enzyme's shape." In which section does this sentence belong?

- A. Results.
- B. Only the Abstract, because explanations never appear in the body of a report.
- C. Discussion.
- D. Methods.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

Which sentence belongs in the Results section of a lab report?

- A. Mean bubble height was 4.2 cm at 37 °C and 0.8 cm at 60 °C.
- B. Heat probably denatured the enzyme at 60 °C.
- C. Future work should test how pH affects the enzyme.
- D. Test tubes were held in water baths for 5 minutes.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A team surveyed 200 ponds in a region known for its hiking trails and found that ponds with more fertilizer runoff had more algae. The data set took two summers to collect. Which conclusion is best supported?

- A.** No conclusion is possible because 200 ponds is too few.
- B.** Runoff and algae are positively associated; a controlled experiment is needed to test whether runoff causes the growth.
- C.** The study proves that runoff causes algae growth.
- D.** Algae growth causes fertilizer runoff to increase.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A team measured the mass of a potato core, which was cut with a 5 mm cork borer, before and after soaking it overnight in distilled water. The mass went from 2.50 g to 3.00 g. The balance reads to 0.01 g. What is the percent change in mass?

- A.** +16.7%
- B.** +0.50%
- C.** -20%
- D.** +20%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

After an antibiotic treatment, a culture's cell count fell by 25% to 1,500 cells/mL. What was the original count?

- A.** 1,875 cells/mL
- B.** 2,000 cells/mL
- C.** 1,125 cells/mL
- D.** 6,000 cells/mL

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A student group ~~working in a well-equipped teaching lab with new digital balances~~ wants to study whether light color affects the growth of radish seedlings. Their instructor, ~~who has taught the course for years~~, asks them to write a hypothesis before ordering seeds. Which statement is a testable hypothesis?

Target word: which statement is a testable hypothesis **Decides:** Only one choice predicts a measurable outcome from a changed factor.

A	Absolute language	"Always ... of any color" cannot be tested by one study and overreaches the design.
B	Correct	It names the changed factor (light color), a measurable outcome (mean height), and a time frame; it can be shown false.
C	Confusion set	That is the research question; a hypothesis is the testable answer to it.
D	Adjacent category	"Prefer" is an opinion about the plant, not a measurable prediction.

2. Answer A Concept 2

In a lab ~~run in a room kept at 22 °C~~, a team places equal-sized pieces of potato in salt solutions of 0%, 2%, 4%, and 6%. Each piece soaks for 30 minutes. Using a scale ~~they calibrated that morning~~, the team records each piece's change in mass. What is the dependent variable?

Target word: the dependent variable **Decides:** Change in mass is what the team measures in response.

A	Correct	The dependent variable is the measured response to the changed factor.
B	Adjacent category	Soaking time is held the same for every piece — a controlled variable.
C	Camouflage	Room temperature is background detail held constant; it is not measured as the outcome.
D	Inverted operation	Salt concentration is what the team changed on purpose — the independent variable.

3. Answer A Concept 2 · transfer

A researcher measures how fast yeast produces carbon dioxide at 20, 30, and 40 °C, using the same amount of sugar and yeast in each flask. What is the independent variable?

Target word: the independent variable **Decides:** Temperature is the factor set at different levels on purpose.

A	Correct	Temperature is deliberately set to three levels.
B	Confusion set	The organism is the subject of the study, not a variable that is changed.
C	Adjacent category	Sugar is held the same — a controlled variable.
D	Inverted operation	That is the measured outcome — the dependent variable.

4. Answer C Concept 3

A lab team tests whether a new liquid fertilizer increases bean plant mass. They buy seeds from a well-known supplier and grow 20 plants in each group. The treatment group gets fertilizer dissolved in 50 mL of water every 3 days. ~~The team's lab notebook is neat and fully dated.~~ What should the control group receive?

Target word: what the control group receives **Decides:** Fertilizer is delivered in 50 mL of water every 3 days.

A	Camouflage	Reacts to the supplier detail; changing seed source adds a new variable.
B	Adjacent category	That is another treatment level, not a control.
C	Correct	The control matches everything except the fertilizer, so the water itself is not a second difference.
D	Dropped element	Drops the water the treatment group receives, so the groups differ in two ways.

5. Answer D Concept 3 • transfer

A student tests an antibacterial soap. She washes one set of countertop swab plates with soap and warm water and leaves a second set completely unwashed. The soap group grows fewer colonies. Why can't she conclude the soap itself reduced bacteria?

Target word: why the conclusion fails **Decides:** The groups differ in soap AND in washing with water.

A	Confusion set	Colony counts are a standard measure; the flaw is the design, not the measure.
B	Absolute language	Negative controls are standard; the problem is this one does not match the treatment.
C	Inverted operation	Soap is the changed factor (independent); colonies are the measured outcome.
D	Correct	A proper control would be washed with warm water only.

6. Answer D Concept 4

A student ~~with an interest in agriculture~~ tests a plant hormone using one treated tomato plant and one untreated tomato plant on a sunny windowsill. After three weeks the treated plant is 12 cm taller. ~~The student's poster uses clear, colorful graphs.~~ What is the most serious weakness of this study?

Target word: most serious weakness **Decides:** One plant per group.

A	Camouflage	Poster design does not decide scientific validity.
B	Absolute language	Both plants share the windowsill; the overreach ignores that it is a shared condition.
C	Wrong stem number	The size of the difference is not the problem; with $n = 1$ any size is unreliable.
D	Correct	Without replication, natural plant-to-plant variation cannot be separated from a hormone effect.

7. Answer C Concept 5

A research assistant ~~in a grant-funded lab~~ records enzyme activity in 10 trials. Nine trials fall between 42 and 48 units; one reads 71 units. ~~The lab director hopes to publish soon, and the deadline is Friday.~~ No exclusion rule was written in the protocol. What should the assistant do?

Target word: proper handling of an unexpected value **Decides:** No exclusion rule was set in advance.

A	Inverted operation	Flips the problem: it cherry-picks the outlier instead of the full data set.
B	Camouflage	Responds to the deadline; timing does not change what must be reported.
C	Correct	Without a pre-set rule, removing data is falsification; the outlier is disclosed and investigated.
D	Dropped element	Drops a real data point without a pre-set rule — falsification.

8. Answer C Concept 6

A simulated lab report on catalase activity is ~~6 pages long and was written by a team of four~~. It contains this sentence: "Enzyme activity likely fell at 60 °C because the high heat changed the enzyme's shape." In which section does this sentence belong?

Target word: correct section **Decides:** The sentence explains why — interpretation.

A	Adjacent category	Results report what was observed, not why.
B	Absolute language	Explanations are the core of the Discussion; "never" is false.
C	Correct	Explaining the cause of a result and linking it to theory is interpretation, which belongs in the Discussion.
D	Confusion set	Methods describe procedures, not explanations of outcomes.

9. Answer A Concept 6 • transfer

Which sentence belongs in the Results section of a lab report?

Target word: which sentence is a Result **Decides:** A Result reports observed data without interpretation.

A	Correct	It states observed data with no explanation.
B	Inverted operation	That interprets the data — Discussion.
C	Adjacent category	Proposing new research belongs in the Discussion.
D	Confusion set	That is a procedure — Methods.

10. Answer B Concept 7

A team surveyed 200 ponds ~~in a region known for its hiking trails~~ and found that ponds with more fertilizer runoff had more algae. ~~The data set took two summers to collect.~~ Which conclusion is best supported?

Target word: best-supported conclusion **Decides:** The data are observational (a survey), not an experiment.

A	Wrong stem number	200 sites is a substantial sample; the limit is the design, not the count.
B	Correct	A survey shows association; cause needs manipulation with other factors controlled.
C	Absolute language	"Proves" overstates any single study, and observational data cannot show cause alone.
D	Inverted operation	Reverses the direction and still reads cause from a correlation.

11. Answer D Concept 8

A team measured the mass of a potato core, ~~which was cut with a 5 mm cork borer~~, before and after soaking it overnight in distilled water. The mass went from 2.50 g to 3.00 g. ~~The balance reads to 0.01 g.~~ What is the percent change in mass?

Target word: percent change **Decides:** $(3.00 - 2.50) \div 2.50 \times 100$.

A	Wrong stem number	Divides by the new mass ($0.50 \div 3.00$) instead of the original.
B	Dropped element	Reports the 0.50 g difference as a percent; drops the divide-by-original step.
C	Inverted operation	Flips the sign; the mass rose.
D	Correct	$0.50 \div 2.50 = 0.20 = +20\%$.

12. Answer B Concept 8 · transfer

After an antibiotic treatment, a culture's cell count fell by 25% to 1,500 cells/mL. What was the original count?

Target word: original value from a percent decrease **Decides:** $\text{New} = \text{original} \times 0.75$.

A	Wrong stem number	Adds 25% of the new count ($1,500 \times 1.25$); the 25% is of the original.
B	Correct	$1,500 \div 0.75 = 2,000$; a 25% drop from 2,000 is 500, leaving 1,500.
C	Inverted operation	Applies the decrease again instead of undoing it.
D	Dropped element	Divides by 0.25, dropping the "1 –" in $1 - 0.25$.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owled.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	4	
Wrong stem number	A figure copied from the prompt that is not the answer.	4	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	8	
Absolute language	Always, never, guarantee — where judgment applies.	5	
Confusion set	Every option is real; the test is exact role-matching.	5	
Camouflage	Answers a stem fact that was never the question.	4	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Testable hypothesis	A hypothesis is a testable, falsifiable prediction that links a changed factor to a measurable outcome.	Treating a research question or an opinion (“plants like blue light”) as a hypothesis.	Answer choices that differ in whether the outcome can actually be measured.
2	Independent, dependent, and controlled variables	The independent variable is changed on purpose, the dependent variable is measured in response, and controlled variables are held the same.	Calling the measured outcome the independent variable, or calling a held-constant factor the thing being tested.	“was changed” vs. “was measured” vs. “was kept the same.”
3	Control groups	A control group gets everything the treatment group gets except the treatment, so any difference can be traced to the treatment.	Thinking the control group should get nothing at all, not even the carrier (water, solvent, handling).	Two groups that differ in more than one way.
4	Replication and sample size	Multiple subjects per group and repeated trials separate a real effect from chance variation; one subject per group supports no general conclusion.	Trusting a big difference from a single trial.	“one plant per group” or a single run.
5	Data integrity	All data collected under the protocol are reported; an outlier may be excluded only by a rule set in advance and disclosed.	Deleting a value because it “must be wrong” or because it hurts the result.	A result that does not fit and a choice about what to do with it.
6	Parts of a lab report	Methods say what was done, Results report what was observed without interpretation, and Discussion explains, compares to the hypothesis, names error sources, and proposes next steps.	Putting explanations in Results or procedures in Discussion.	A sentence that explains “why” vs. one that reports “what.”
7	Support, not proof; association, not cause	Data support or fail to support a hypothesis; one study never proves it, and an observed correlation does not by itself show cause.	Writing “the data prove” or reading cause from a survey.	Observational data paired with a causal-sounding conclusion.
8	Percent change	Percent change = $(\text{new} - \text{original}) \div \text{original} \times 100$.	Dividing by the new value or reporting the raw difference as a percent.	“by what percent did it change.”

Course scope (public description): This course focuses on developing foundational skills in scientific investigation within the field of biology. It emphasizes the application of the scientific method to answer biological questions through hypothesis-driven experimentation. Students will learn to design, execute, and analyze biological experiments, ensuring adherence to rigorous scientific protocols and ethical standards. The course also covers essential aspects of identifying and analyzing the components of a lab report by answering questions related to a simulated lab report. Throughout the course, students will cultivate critical thinking skills necessary for interpreting data, drawing conclusions, and proposing further research directions in biology.